CS260 | Data Structures
Reading Assignment - Week 5

1 Instructions

- This assignment is worth 5 points.
- All the following readings are to be done from the book *Algorithms Illuminated by Tim Roughgarden*
- From *Chapter 10: The Heap Data Structure*, read *section 10.1 (Data Structures: An overview)*, *section 10.2 (Supported Operations)*, *section 10.3 (Applications)*, and skim *section 10.5 (Implementation Details)*
- **Take brief notes while reading**. Write your notes on a sheet of paper or on your computer.
- **Answer the questions below**. Write your answers on a sheet of paper or on your computer.
- **Submission:** Take a picture of your notes and answers (or a screenshot, or a pdf of the notes on computer), and upload it to gradescope.
- **Grading:** Grading will be based on effort, not correctness.

2 Questions

Question 1 : 2 points

Provide the notes you took for the readings.

Question 2 : 3 points

(a) [1 point] How does principle of parsimony apply to algorithm design when it comes to choosing a data structure?

(b) [1 point] You are being asked to keep track of the median value of some input and print it out efficiently when asked. Since you don't know when you will be asked to tell the median number, an efficient method is to keep track of the median element using two heaps, as explained in section 10.3.3. Getting to know the median element in this case will lead to $\log i$ time by calling extract max/min functions once, if the mean is asked when the $2i$th element is inserted. Although this is efficient, this is just part of the story.

If the total input is consisting of n elements, what is the total cost of maintaining the two heaps?

(c) [1 point] As mentioned in Section 10.5, both ExtractMin and Insert functions are $\log n$ time operations. Consider a min heap where we would like to update the value of a key to a value that is lesser. How would you achieve this operation and what would be the runtime of this operation?